A First Course in Abstract Algebra

Section 26: Homomorphisms and Factor Rings

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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26 Homomorphisms and Factor Rings

Exercise 26.1: Homomorphisms of the product of integer rings

Plan. A homomorphism is determined by $u = \phi(1,0)$ and $v = \phi(0,1)$, since $\phi(m,n) = mu + nv$. The identities $(1,0)^2 = (1,0)$, $(0,1)^2 = (0,1)$ and $(1,0)(0,1) = 0$ force $u^2 = u$, $v^2 = v$, and $uv = 0$. An integer idempotent satisfies $t(t-1) = 0$, so each coordinate of u, v is 0 or 1. In each output coordinate, the pair of coefficients is therefore $(0,0)$, $(1,0)$, or $(0,1)$. Consequently each of the two coordinates of $\phi(m,n)$ can independently be 0, m, or n: there are $3^2 = 9$ maps. Conversely these nine maps preserve coordinatewise sums and products, so the list is exhaustive. Homomorphisms here are not required to preserve the unity of the codomain.

Exercise 26.2: A two-element subring

A subring isomorphic to $\mathbb{Z}_2$ must be $\{0, e\}$ with $e \neq 0$, $2e = 0$, and $e^2 = e$ modulo n. The additive cyclic group $\mathbb{Z}_n$ has an element of order two exactly when $n = 2m$, and then its only such element is $e = m$. Now $m^2 \equiv m \pmod{2m}$ is equivalent to $2 \mid (m-1)$. Thus m must be odd.

Conclusion. Exactly $n \equiv 2 \pmod 4$ work, with subring $\{0, n/2\}$. Its nonzero element acts as the identity within that subring.

Exercise 26.3: Ideals and quotients of the twelve-element ring

Every additive subgroup of $\mathbb{Z}_{12}$ is $\langle d \rangle$ for a divisor d of 12, and it is an ideal since multiplying a multiple of d by any residue still gives a multiple of d. Reduction modulo d has kernel $\langle d \rangle$, so

N	$\mathbb{Z}_{12}/N$
$\langle 1 \rangle = \mathbb{Z}_{12}$	$\{0\}$
$\langle 2 \rangle = \{0,2,4,6,8,10\}$	$\mathbb{Z}_2$
$\langle 3 \rangle = \{0,3,6,9\}$	$\mathbb{Z}_3$
$\langle 4 \rangle = \{0,4,8\}$	$\mathbb{Z}_4$
$\langle 6 \rangle = \{0,6\}$	$\mathbb{Z}_6$
$\langle 12 \rangle = \{0\}$	$\mathbb{Z}_{12}$

These are all the ideals because an ideal must first be an additive subgroup.

Exercise 26.4: Tables for an even-integer quotient

Write $A = 8\mathbb{Z}$, $B = 2 + 8\mathbb{Z}$, $C = 4 + 8\mathbb{Z}$, and $D = 6 + 8\mathbb{Z}$. Add or multiply representatives and then reduce modulo 8:

$+$	A	B	C	D
A	A	B	C	D
B	B	C	D	A
C	C	D	A	B
D	D	A	B	C

$\cdot$	A	B	C	D
A	A	A	A	A
B	A	C	A	C
C	A	A	A	A
D	A	C	A	C

For example $BD = 12 + 8\mathbb{Z} = C$. Every element squares to A or C, and no element is a multiplicative identity: $uC = A \neq C$ for all u. Since $\mathbb{Z}_4$ has an identity, these rings are not isomorphic, although their additive groups are isomorphic.

Exercise 26.5: Correcting isomorphism

The condition $\operatorname{Ker}\phi = \{0\}$ gives injectivity, but not surjectivity onto the declared codomain. An isomorphism $R \to R'$ is a *bijective* ring homomorphism. For instance inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ has zero kernel and is not an isomorphism onto $\mathbb{Q}$.

Exercise 26.6: The ideal definition

The definition is correct: an ideal is an additive subgroup N with $rn, nr \in N$ for every $r \in R, n \in N$. Both absorption conditions are needed in a noncommutative ring; they coincide when multiplication is commutative.

Exercise 26.7: Correcting the kernel

The kernel consists of *inputs*, not equations or outputs:

$$\operatorname{Ker}\phi = \{r \in R : \phi(r) = 0_{R'}\}.$$

For example the kernel of reduction $\mathbb{Z} \to \mathbb{Z}_5$ is $5\mathbb{Z}$, a subset of $\mathbb{Z}$.

Exercise 26.8: Differentiation

Differentiation preserves addition, but the product rule is $\delta(fg) = f'g + fg'$, rather than $f'g'$. Take $f = g = x$: $\delta(x^2) = 2x$ and $\delta(x)\delta(x) = 1$. Thus δ is not a ring homomorphism. Its kernel is the subring of constant functions (a differentiable function with derivative zero is constant). Example 26.12 shows this subring is not an ideal; this is another obstruction, since kernels of ring homomorphisms must be ideals.

Exercise 26.9: An image identity need not be the codomain identity

Let $\phi : \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be $\phi(n) = (n, 0)$. Coordinatewise calculation gives $\phi(m + n) = \phi(m) + \phi(n)$ and $\phi(mn) = \phi(m)\phi(n)$. Yet $\phi(1) = (1, 0)$ is nonzero and differs from the codomain identity $(1, 1)$. It is the identity only on the image.

Exercise 26.10: True and false statements

(a) True. A homomorphism induces $R/\operatorname{Ker}\phi \cong \phi[R]$.

(b) False. Inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ sends the ideal $\mathbb{Z}$ to $\mathbb{Z}$, which is not an ideal of $\mathbb{Q}$, since $(1/2)1 \notin \mathbb{Z}$.

(c) True. $\phi(r) = \phi(s)$ exactly when $r - s \in \operatorname{Ker}\phi$.

(d) False. $1 \in \mathbb{Q}$ but $\sqrt{2}\,1 \notin \mathbb{Q}$, so absorption in $\mathbb{R}$ fails.

(e) True. Additive closure is required, and ideal absorption includes closure under multiplication of its own elements.

(f) False. $\mathbb{Z} \subset \mathbb{Q}$ is the counterexample from (b).

(g) True. $(r+N)(s+N) = rs+N = sr+N$ if R is commutative.

(h) True. Reduction modulo four induces $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4$.

(i) True. If $1 \in N$, every $r = r1 \in N$; conversely $N = R$ contains 1.

(j) True. These are precisely the respective substructures allowing well-defined quotient operations.

Exercise 26.11: The extreme quotients

The ring R/R has just the one coset R and is the zero ring. The map $r + \{0\} \mapsto r$ identifies $R/\{0\}$ with R. Thus the two extreme quotients contain no new structure, although both are useful boundary cases in general theorems.

Exercise 26.12: A domain with a field quotient

Take the integral domain $\mathbb{Z}$ and its ideal $2\mathbb{Z}$. Reduction modulo two gives $\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2$, which is a field because its only nonzero element is its own inverse.

Exercise 26.13: A domain with zero divisors in a quotient

Again start with $\mathbb{Z}$, but quotient by $6\mathbb{Z}$. In $\mathbb{Z}/6\mathbb{Z}$, the cosets $2+6\mathbb{Z}$ and $3+6\mathbb{Z}$ are nonzero and their product is $6+6\mathbb{Z} = 0+6\mathbb{Z}$. Thus an integral domain can have a quotient with zero divisors.

Exercise 26.14: Removing zero divisors by a quotient

The ring $R = \mathbb{Z} \times \mathbb{Z}$ has $(1,0)(0,1) = 0$ with both factors nonzero. Projection $(a,b) \mapsto a$ has ideal kernel $N = \{0\} \times \mathbb{Z}$ and gives $R/N \cong \mathbb{Z}$, an integral domain.

Exercise 26.15: A subring that is not an ideal

The diagonal $D = \{(n,n) : n \in \mathbb{Z}\}$ is closed under differences and products and contains zero. But $(1,0)(1,1) = (1,0) \notin D$, so it is not an ideal of $\mathbb{Z} \times \mathbb{Z}$.

Exercise 26.16: Representatives and quotient elements

(a) The student has used r, s for elements of R/N and then written $rs - sr \in N$, which should concern elements of R. Quotient elements are cosets, so the types of objects have been confused. **(b)** A correct start is: for all $r, s \in R$, commutativity gives $(r+N)(s+N) = (s+N)(r+N)$. **(c)** That equality is equivalent to $rs+N = sr+N$, which is equivalent to $rs - sr \in N$. Reading these equivalences in either direction proves both implications. Notice that $rs = sr$ in R itself is not required.

Exercise 26.17: A quadratic integer ring and matrices

Plan. Match the coefficients of 1 and $\sqrt{2}$ with matrix entries. If $u = a + b\sqrt{2}$ and $v = c + d\sqrt{2}$, then

$$u - v = (a-c) + (b-d)\sqrt{2}, \qquad uv = (ac+2bd) + (ad+bc)\sqrt{2}.$$

These formulas show R is a nonempty subring of $\mathbb{R}$. Set $M(a,b) = \begin{pmatrix} a & 2b \\ b & a \end{pmatrix}$. Matrix subtraction and multiplication give

$$M(a,b) - M(c,d) = M(a-c, b-d), \quad M(a,b)M(c,d) = M(ac+2bd, ad+bc),$$

so R' is a subring of $M_2(\mathbb{Z})$ and $\phi(a+b\sqrt{2}) = M(a,b)$ preserves the operations. Coefficients in $a+b\sqrt{2}$ are unique: otherwise irrationality of $\sqrt{2}$ would fail. Hence the map is well defined; its kernel is zero by its entries, and every displayed matrix is an image. Thus it is a ring isomorphism.

Exercise 26.18: Homomorphisms from a field

Let $\phi : F \to R$ have a nonzero kernel element a. For any $x \in F$, write $x = (xa^{-1})a$. Then $\phi(x) = \phi(xa^{-1})\phi(a) = 0$, so ϕ is the zero map. If no such a exists, the kernel is zero, and ϕ is injective. These are the only possibilities.

Exercise 26.19: Composing ring homomorphisms

For $a, b \in R$, apply the defining identities twice:

$$\psi\phi(a+b) = \psi(\phi(a) + \phi(b)) = \psi\phi(a) + \psi\phi(b),$$
$$\psi\phi(ab) = \psi(\phi(a)\phi(b)) = \psi\phi(a)\,\psi\phi(b).$$

Thus $\psi \circ \phi : R \to R''$ preserves both operations and is a ring homomorphism.

Exercise 26.20: Frobenius

In a commutative ring the binomial theorem applies. For prime p, each $\binom{p}{k}$ with $0 < k < p$ is divisible by p, so characteristic p gives

$$(a+b)^p = a^p + b^p.$$

Commutativity also gives $(ab)^p = a^p b^p$. Therefore $a \mapsto a^p$ preserves addition and multiplication and is a ring homomorphism. Commutativity is essential to these expansions.

Exercise 26.21: The image of the identity in a ring without zero divisors

Put $e = \phi(1)$. It satisfies $e^2 = e$. It is nonzero, because otherwise $\phi(r) = \phi(1r) = e\phi(r) = 0$ for every r, contrary to the nonzero-image assumption. For any $y \in R'$, we have $e(ey - y) = 0$. Since $e \neq 0$ and R' has no zero divisors, $ey = y$. Likewise $(ye - y)e = 0$ gives $ye = y$. Thus e is a two-sided identity for all of R', even when ϕ is not onto.

Exercise 26.22: Images and inverse images of ideals

(a) The image $\phi[N]$ is an additive subgroup. For $\phi(r) \in \phi[R]$ and $\phi(n) \in \phi[N]$, both products equal images of rn or nr in N, so absorption holds inside $\phi[R]$. **(b)** Inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ with $N = \mathbb{Z}$ fails absorption by $1/2$, giving the required counterexample. **(c)** Let $I = \phi^{-1}[N']$. Then $0 \in I$, and for $a, b \in I$, $\phi(a-b) = \phi(a) - \phi(b) \in N'$. For $r \in R$, both $\phi(ra) = \phi(r)\phi(a)$ and $\phi(ar) = \phi(a)\phi(r)$ lie in N'. Thus I is an ideal. This uses only absorption by elements in $\phi[R]$, so either stated choice of ambient ring for N' is sufficient.

Exercise 26.23: Polynomials vanishing on a set

Let N_S be the polynomials vanishing at each point of $S \subset F^n$. The zero polynomial belongs to N_S. If $f, g \in N_S$, then $(f-g)(a) = f(a) - g(a) = 0$ for every $a \in S$. If h is any polynomial, $(hf)(a) = h(a)f(a) = 0$ and similarly for fh. Thus the subring test and absorption show that N_S is an ideal. If S is empty, the condition is vacuous and $N_S = F[x_1, \ldots, x_n]$, still an ideal.

Exercise 26.24: Quotients of a field

An ideal N of a field F is either zero or all of F: if $0 \neq a \in N$, then $1 = a^{-1}a \in N$, and hence every $x = x1 \in N$. In the two cases, F/N is respectively isomorphic to F or to the one-element zero ring.

Exercise 26.25: Unity in a proper quotient

The coset $1 + N$ multiplies any $r + N$ to $r + N$ on either side. It is nonzero because $1 \in N$ would force $N = R$, contrary to the assumption. Thus a proper quotient of a ring with unity again has a nonzero unity.

Exercise 26.26: An annihilator ideal

Let $I_a = \{x : ax = 0\}$. It contains zero. If $x, y \in I_a$, then $a(x-y) = 0 - 0 = 0$, so it is an additive subgroup. For $r \in R$, commutativity gives $a(rx) = r(ax) = 0$, hence $rx = xr \in I_a$. Thus I_a is an ideal. Without commutativity, a one-sided statement is automatic but both sides need not follow.

Exercise 26.27: Intersections of ideals

For a family (I_j) of ideals, zero belongs to every I_j. If a, b belong to their intersection, $a - b \in I_j$ for every j, hence lies in the intersection. Likewise ra and ar belong to every I_j for every $r \in R$. This proves the ideal axioms. The intersection of an empty family is understood as R, also an ideal.

Exercise 26.28: Induced quotient homomorphism

Define $\overline{\phi}(r + N) = \phi(r) + N'$. If $r + N = s + N$, then $r - s \in N$ and $\phi(r) - \phi(s) \in \phi[N] \subset N'$, so the output cosets agree. Thus the formula is well defined. Moreover

$$\overline{\phi}((r + N) + (s + N)) = \phi(r + s) + N' = \overline{\phi}(r + N) + \overline{\phi}(s + N),$$

and the identical calculation using products proves multiplicativity. No surjectivity assumption is needed.

Exercise 26.29: Images of units under a surjection

Because ϕ is onto, $\phi(1)$ acts as a two-sided identity on every element of R': write it as $\phi(r)$ and use $1r = r1 = r$. Also $\phi(1) \neq 0$ since $R' \neq \{0\}$. If $uv = vu = 1$ in R, applying ϕ gives $\phi(u)\phi(v) = \phi(v)\phi(u) = 1_{R'}$. Thus $\phi(u)$ is a unit with inverse $\phi(v)$.

Exercise 26.30: The nilradical

Let N consist of nilpotent elements in a commutative ring. It contains zero. If $a^m = 0$ and $b^n = 0$, expand $(a-b)^{m+n-1}$. Each term contains either at least m factors a or at least n factors b, so every term is zero. Thus $a - b \in N$. For $r \in R$, $(ra)^m = r^m a^m = 0$, so $ra = ar \in N$. These facts prove that N is an ideal. Commutativity is used in both power calculations.

Exercise 26.31: Three nilradicals

In $\mathbb{Z}_{12}$ a nilpotent residue must be divisible by both primes $2, 3$, since reducing a vanishing power modulo either prime gives a vanishing element in a field. Conversely $6^2 \equiv 0 \pmod{12}$. Thus the nilradical is $\{0, 6\} = \langle 6 \rangle$. In $\mathbb{Z}$, a vanishing integer power forces the integer to be zero, so the nilradical is $\{0\}$. In $\mathbb{Z}_{32}$, precisely the even residues are nilpotent: $(2k)^5 \equiv 0$, while an odd residue is a unit and cannot have a zero power. Thus the nilradical is $\langle 2 \rangle$.

Exercise 26.32: Quotienting by all nilpotents

Suppose $(a + N)^m = N$ in R/N. Then $a^m \in N$, and because N is the nilradical, $(a^m)^n = 0$ for some n. Hence $a^{mn} = 0$, so $a \in N$ and $a + N = N$. Thus the quotient has no nonzero nilpotent elements.

Exercise 26.33: Lifting nilpotence

For any $a \in R$, the hypothesis that every coset is nilpotent gives $(a + N)^m = N$, hence $a^m \in N$. Every element of N is nilpotent, so $(a^m)^n = 0$ for some n. Thus every $a \in R$ is nilpotent, which says the nilradical is all of R. The exponents may depend on a.

Exercise 26.34: The radical of an ideal

Let $\sqrt{N} = \{a : a^m \in N \text{ for some } m \geq 1\}$. It contains zero. If $a^m, b^n \in N$, expand $(a-b)^{m+n-1}$; each term has a factor a^m or b^n, so each term belongs to N. This puts $a - b$ in $\sqrt{N}$. For any $r \in R$, $(ra)^m = r^m a^m \in N$, so $ra = ar \in \sqrt{N}$. Therefore $\sqrt{N}$ is an ideal. This proof does not require all elements of N themselves to be nilpotent.

Exercise 26.35: A radical can grow or remain the same

(a) In $\mathbb{Z}$, take $N = 4\mathbb{Z}$. Then $2^2 \in 4\mathbb{Z}$ but $2 \notin 4\mathbb{Z}$, so $\sqrt{N} \neq N$ (in fact $\sqrt{N} = 2\mathbb{Z}$). **(b)** Take $N = 2\mathbb{Z}$. If a^m is even then a is even, so $\sqrt{2\mathbb{Z}} = 2\mathbb{Z}$. Both examples use proper ideals.

Exercise 26.36: Radical versus quotient nilradical

An element of the nilradical of R/N is a *coset* $a + N$ such that $(a + N)^m = N$, equivalently $a^m \in N$. Hence

$$\mathrm{Nil}(R/N) = \{a + N : a \in \sqrt{N}\} = \sqrt{N}/N.$$

It is not literally $\sqrt{N}$, whose elements lie in R rather than R/N.

Exercise 26.37: Complex numbers as real matrices

Set $M(a,b) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Direct multiplication gives

$$M(a,b)M(c,d) = M(ac - bd, ad + bc),$$

which matches $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$. Addition also matches entry by entry. The map is injective because $M(a,b) = 0$ forces $a = b = 0$, and it is onto its image by definition. Therefore $\mathbb{C} \cong \phi[\mathbb{C}] \subset M_2(\mathbb{R})$. The sign convention corresponds to multiplication on coordinates $(x, -y)$ instead of (x, y) and is consistent.

Exercise 26.38: A ring version of the regular representation

(a) Left multiplication λ_a is additive because $a(x+y) = ax + ay$, so it is an endomorphism of $(R, +)$.
(b) Endomorphisms are added pointwise and multiplied by composition. Thus

$$\lambda_a - \lambda_b = \lambda_{a-b}, \qquad \lambda_a \circ \lambda_b = \lambda_{ab}.$$

Together with $\lambda_0 = 0$, these identities show that $R' = \{\lambda_a : a \in R\}$ is a subring. **(c)** The map $a \mapsto \lambda_a$ is a surjective ring homomorphism onto R'. If $\lambda_a = \lambda_b$, evaluate at 1 to obtain $a = b$. Thus it is injective and $R \cong R'$. Unity is what makes this final injectivity argument work.